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Preliminary palaeomagnetic results of an Archaean dolerite dyke of west Greenland: geomagnetic field intensity at 2.8 Ga

Chiyo Morimoto,[1] Yo-ichiro Otofuji,[1] Masako Miki,[1] Hidefumi Tanaka[2,*] and Tetsumaru Itaya[3]

[1] *Department of Earth and Planetary Sciences, Faculty of Science, Kobe University, Kobe, Japan*
[2] *Department of Applied Physics, Tokyo Institute of Technology, Meguro-ku, Tokyo, Japan*
[3] *Hiruzen Research Institute Okayama Branch, Okayama University of Science, Okayama, Japan*

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SUMMARY

The geomagnetic field intensity during Archaean times is evaluated from a palaeomagnetic and chronological study of a dolerite dyke intruded into the 3000 Ma Nuuk Gneisses at Nuuk (64.2°N, 51.7°W), west Greenland. Plagioclase from the dolerite dyke yields a mean K–Ar age of 2752 Ma. Palaeomagnetic directions after thermal demagnetization of the dyke and the gneiss reveal a positive baked-contact test, indicating that the high-temperature-component magnetization of the dyke is primary. Thellier experiments on 12 dyke specimens yield a palaeointensity value of 13.5 ± 4.4 μT. The virtual dipole moment at *ca.* 2.8 Ga is $1.9 \pm 0.6 \times 10^{22}$ Am^2, which is about one-quarter of the present value. The present study and other available data imply that the Earth's magnetic field at $2.7 \sim 2.8$ Ga was characterized by a weak dipole moment and that a fairly strong geomagnetic field similar to the present intensity followed the weak field after *ca.* 2.6 Ga.

Key words: Archaean, Earth's core, geomagnetic field, Greenland, K–Ar age, palaeomagnetism.

INTRODUCTION

The magnetic field behaviour of the early Earth provides significant information about the evolution of the Earth's core. The timing of separation from a primitive core to the present stratified core might be estimated through the appearance of a fairly strong geomagnetic field on a billion-year timescale in terms of the energetics of the core (Stevenson, Spohn & Shubert 1983). Palaeo-geomagnetic intensity data compiled by Hale (1987a) from Archaean rocks of the African and Canadian shields with ages between 3.5 Ga and 0.8 Ga suggest that a marked increase in magnitude of the geomagnetic field occurred between 2.7 Ga and 2.1 Ga. A recent compilation of the most reliable palaeointensity data by Prevot & Perrin (1992), however, hardly recognizes this palaeointensity increase. This is because of a shortage of data within this interval. The accumulation of more palaeointensity data from Archean times is therefore required to show the evolution of the core.

The present study aims to investigate the palaeointensity for an Archaean dolerite dyke in west Greenland (Fig. 1). The Archaean gneiss complex, with radiometric ages between 3800 Ma and 2900 Ma, is exposed in the southwestern part of Greenland (Escher & Watt 1976; Nutman *et al.* 1989; McGregor, Friend & Nutman 1991). Major dolerite swarms were emplaced into the Archaean gneiss after the end of plutonic activity at about 2700–2500 Ma (Nielsen 1985). The dolerites are often well preserved, without obvious metamorphic effects although the dykes have been named metadolerite dykes (MD). The fresh dolerite dyke in the MD is our target for this study.

The Nuuk region of west Greenland is characterized by the Nuuk quartzofeldspathic gneisses of *ca.* 3000 Ma (Pankhurst, Moorbath & McGregor 1973; Taylor *et al.* 1980; Baadsgaard & McGregor 1981; McGregor *et al.* 1991). The dolerite dyke intrudes the Nuuk gneisses at Nuuk town (64.2°N, 51.7°W) (Fig. 1). It trends approximately 110° and has a width of 4 m. 32 block samples oriented with a sun compass were collected along a profile roughly at right angles to the dyke contact: 13 from the dyke and 19 from the Nuuk gneisses over a distance ranging from 3 cm to 55 m from the contact. Another block sample was collected from the dolerite dyke for K–Ar dating.

K–Ar DATING

Age determination was carried out on plagioclase separates of the dolerite dyke (Table 1). Prior to K–Ar dating, the dolerite

* Present address: Faculty of Education, Kochi University, Kochi 780, Japan.

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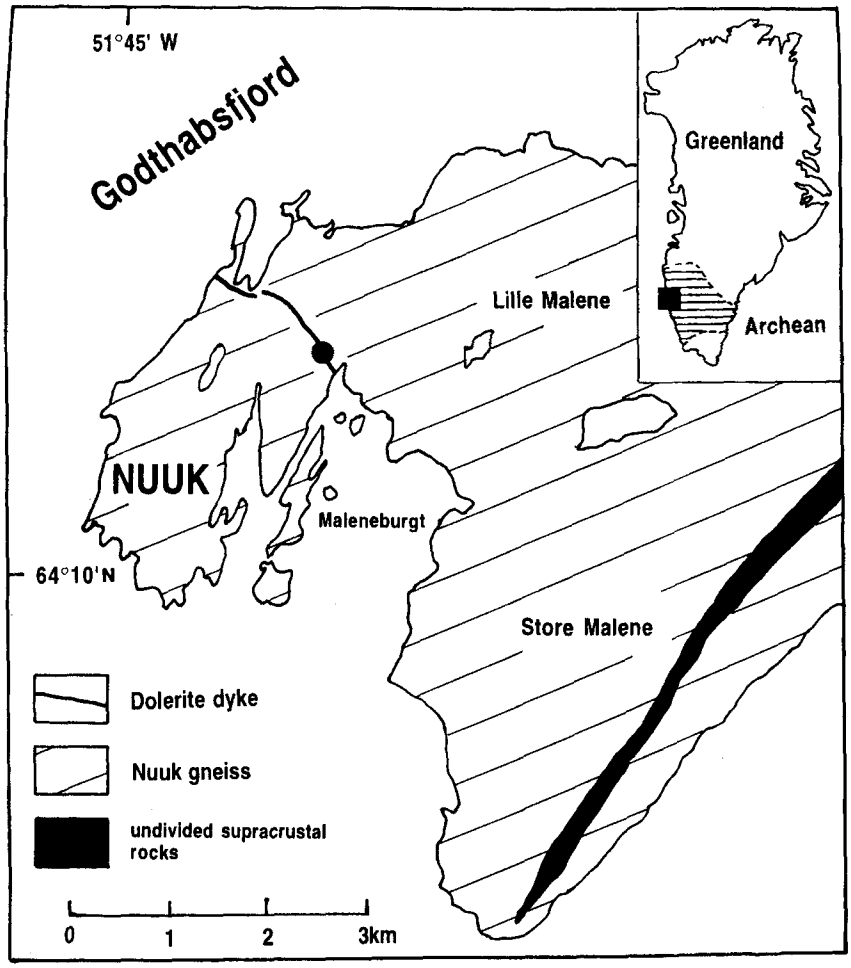


Figure 1. Index map of Greenland, with a simplified geological map of the Nuuk area after McGregor *et al.* (1991). The solid circle indicates the sampling point of the dolerite dyke and the Nuuk gneisses.

Table 1. K–Ar age determinations for the dolerite dyke, Nuuk area. Ages are calculated from the following decay and abundance constants: $\lambda_\varepsilon = 0.581 \times 10^{-10}\ yr^{-1}$; $\lambda_\beta = 4.962 \times 10^{-10}\ yr^{-1}$; $^{40}K/K = 1.167 \times 10^{-4}$ mol mol^{-1} (Steiger & Jäger 1977).

Sample	K–Ar age (Ma)	Potassium (wt. per cent)	Rad. argon 40 (10-8ccSTP/g)	Non rad. Ar (per cent)
GP01-A	2707 ± 76	0.160 ± 0.008	3899 ± 84	0.9
(Plagioclase)	2796 ± 72	0.160 ± 0.008	4152 ± 41	0.7
mean	2752 ± 63			

sample was investigated by optical microscopy and using an electron microprobe. Although a part of the pyroxene is replaced by hornblende, the igneous fabric is preserved and the plagioclases have survived alteration, indicating that hornblende replacement is due to autometamorphism.

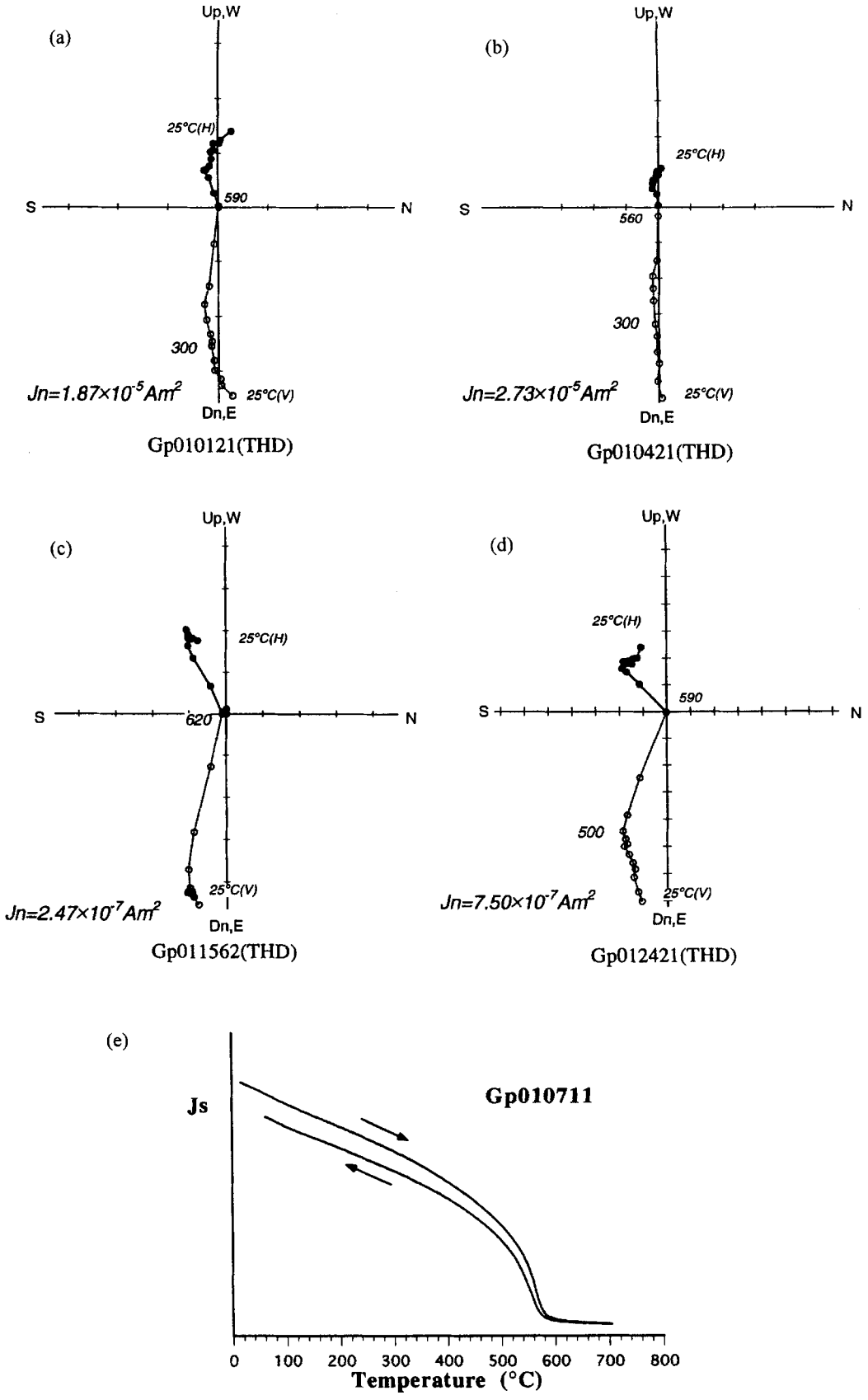
Samples for K–Ar analysis were prepared using the method described by Briggs *et al.* (1989). Plagioclase separates of 80–100 mesh size were used for argon analysis, and a portion of each separate was further ground to use for potassium analysis. Analyses of potassium and argon, and calculations of ages and errors, were carried out according to the method described by Nagao *et al.* (1984) and Itaya *et al.* (1991). The decay constants of Steiger & Jäger (1977) were used in the age calculations.

Ages for two portions of plagioclase separates are given as 2707 ± 76 Ma and 2796 ± 72 Ma (mean: 2752 ± 63 Ma). These two plagioclase ages are statistically indistinguishable, suggesting that the age determination is reproducible. The mean age of 2750 Ma implies that the dyke was intruded into the Nuuk gneisses during collision events of the Akia and Tasiusarsuaq terranes in the Gothabsfjord–Isua region in late Archaean times (McGregor *et al.* 1991).

PALAEOMAGNETIC DIRECTIONS

In the laboratory, two to 28 specimens, 25 mm in diameter and 22 mm long, were prepared from each sample. Natural

Figure 2. Demagnetization behaviour and thermomagnetic analysis. Orthogonal projections of magnetization vector endpoints during thermal demagnetization (THD) experiments for the following specimens: (a), (b) the dolerite dyke, (c) the Nuuk gneisses 6 cm from the contact, and (d) the Nuuk gneisses 28 m from the contact. Open (solid) symbols show the magnetization vector projected onto the vertical (V) [horizontal (H)] plane. Numbers beside symbols represent temperature (°C). Jn: intensity of natural remanent magnetization. (e) High-field saturation magnetization (Js) versus temperature curves for the dyke sample (GP010711). Experiments were performed in air. Field intensity was 0.8 T. The arrows indicate heating and cooling.


(a)
Up,W
25°C(H)
590
S
N
300
Jn=1.87×10⁻⁵Am²
25°C(V)
Dn,E
Gp010121(THD)
(b)
Up,W
25°C(H)
S
N
560
300
Jn=2.73×10⁻⁵Am²
25°C(V)
Dn,E
Gp010421(THD)
(c)
Up,W
25°C(H)
S
N
620
Jn=2.47×10⁻⁷Am²
25°C(V)
Dn,E
Gp011562(THD)
(d)
Up,W
25°C(H)
590
S
N
500
Jn=7.50×10⁻⁷Am²
25°C(V)
Dn,E
Gp012421(THD)
(e)
Js
Gp010711
0
100
200
300
400
500
600
700
800
Temperature (°C)

remanent magnetizations (NRMs) were measured with either a spinner magnetometer (Natsuhara SMM-85) or an ScT cryogenic magnetometer according to the intensity of magnetization. The NRM intensities of the dolerite dyke samples are between 1.5×10^{-5} Am^2 and 3.7×10^{-5} Am^2, and those of the gneiss samples range from 1.6×10^{-7} Am^2 to 2.6×10^{-6} Am^2. At least one specimen from each sample was thermally demagnetized in *ca.* 14 steps to 630 °C (Fig. 2). Alternating field demagnetization proved for the most part to be ineffective. The characteristic direction for the high-temperature component was calculated by principal component analysis (Kirschvink 1980).

Two magnetization components were identified in 30 per cent of the block samples of dolerite and in 60 per cent of the block samples of gneiss (Fig. 2). A low-temperature component was removed in the temperature range 100 to 400 °C for the dolerite and 250 to 560 °C for the gneiss. A high-temperature component with maximum unblocking temperatures near 580 °C was observed for both the dyke and gneiss samples, indicating that this component resides in titanium-free magnetite. This is confirmed by thermomagnetic analysis: the Curie temperature ranges from 577 to 582 °C.

The high-temperature components in all 13 dyke samples showed normal polarity and their directions were tightly clustered (Fig. 3). Similar directions were observed for samples from both marginal and interior parts of the dyke. The mean palaeomagnetic direction for the dyke is $D = 255.4°$, $I = 74.6°$, $\alpha_{95} = 4.7°$. Since the angular distance is $15.0° \pm 4.7°$ between this mean direction and the present geomagnetic field direction ($D = 322.9°$, $I = 79.0°$; Langel 1992), the two directions are statistically distinguished.

A positive baked-contact test for the Nuuk gneisses demonstrates the probable primary nature of the dyke magnetization (Fig. 3). The gneiss is divided into two zones from a palaeomagnetic view-point: a metamorphosed–hybrid zone from the contact to 1.5 m, and a non-metamorphosed zone between 1.5 m and 55 m. The palaeomagnetic direction of the gneiss between 1.5 m and 55 m from the contact is $D = 225.9°$, $I = 55.2°$ and $\alpha_{95} = 4.8°$, which is the same direction as for the Archaean gneiss complex around Nuuk ($D = 227°$, $I = 59°$) reported by Fahrig & Bridgwater (1976). This direction is significantly different from the dyke direction, by $22.6° \pm 6.7°$ (Fig. 3a). The palaeomagnetic direction of the gneiss within 1.5 m from the contact exhibits intermediate palaeomagnetic directions between those of the dolerite and the gneiss far from the contact. The angular difference is $12.4° \pm 9.1°$ for 3 cm from the contact. The angular distance between the direction of a gneiss sample and the mean palaeomagnetic direction of the dyke is plotted as a function of distance for the gneiss sample from the contact (Fig. 3b). The best-fit curve by a power function indicates that the angular distance decreases as the location of the gneiss samples approaches the contact. These observations indicate that the gneiss NRMs within 1.5 m of the dyke contact are partially replaced by dyke thermal overprints, and that the gneiss more than 1.5 m from the contact in an unbaked zone.

There remains significant deflection between the palaeomagnetic directions of the baked gneiss and the dyke. This deflection is attributable to the anisotropic fabric of the gneiss rocks (Stephenson, Sadikun & Potter 1986). Measurement of the magnetic susceptibility anisotropy shows that the maximum axis of the susceptibility ellipsoid for the gneiss specimens

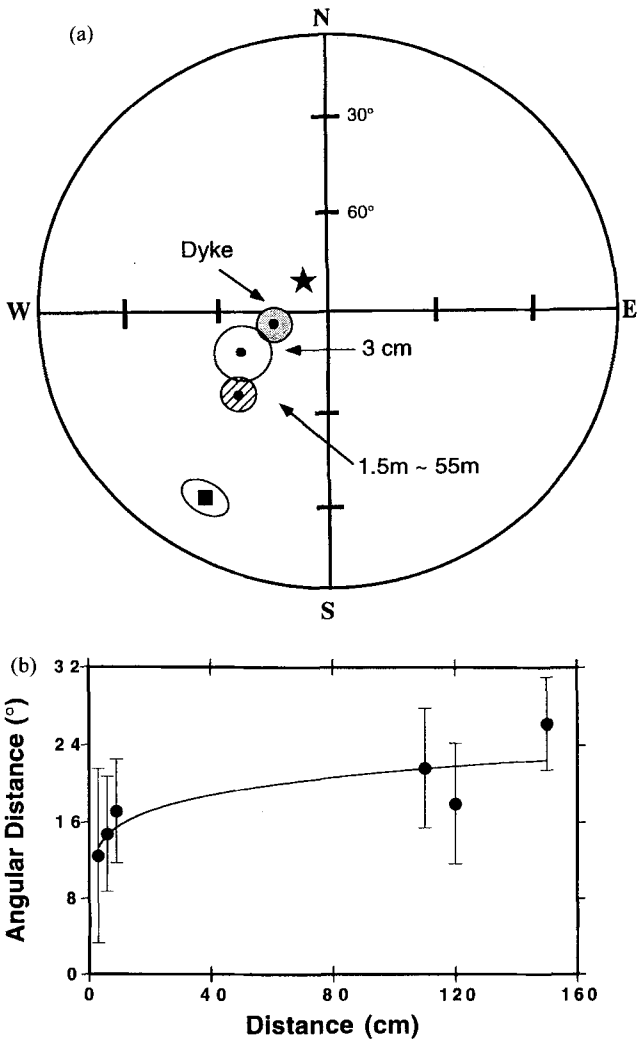


Figure 3. Palaeomagnetic directions and orientation of the mean maximum susceptibility axis. (a) The mean palaeomagnetic direction with its 95 per cent confidence limit for the 13 dolerite dyke samples (shaded). Palaeomagnetic directions with 95 per cent confidence limits are shown also for the Nuuk gneisses 3 cm and 1.5 m ~ 55 m (hatched) away from the contact. The orientation of the mean maximum susceptibility axis of the adjoining Nuuk gneisses (3 cm ~ 55 m) is also plotted as the square with its 95 per cent confidence limit. Susceptibility anisotropy was measured on a spinner magnetometer (Schonstedt SSM-1A). All the directions are plotted on the lower hemisphere. The present Earth field is shown by the star. Equal-area projection. (b) Plot of the angular distance between the palaeomagnetic direction of a gneiss sample and the mean palaeomagnetic direction of the dyke, as a function of distance of the gneiss sample from the contact. The error bars show the uncertainty in the angular distance at the 95 per cent confidence level. The solid line shows the best fit for a power-function curve.

is aligned in the direction $D = 212.3°$, $I = 21.0°$ ($\alpha_{95} = 5.6°$) (Fig. 3), whereas no specific direction is identified in the distribution of the maximum axes of the dyke specimens. The palaeomagnetic direction of the baked gneiss is probably a result of a thermoremanent magnetization (TRM) parallel to the dyke palaeomagnetic direction and a remanent magnetism parallel to the maximum axis of the shape anisotropy (Hyodo & Dunlop 1993).

PALAEOINTENSITY

Palaeomagnetic field intensity was evaluated by Coe's modification (Coe 1967) of the Thellier method (Thellier & Thellier 1959) for at least one specimen for each dyke sample (Fig. 4 and Table 2). Nitrogen gas was run through the furnace for the steps over 150 °C to minimize the oxidation of magnetic minerals. The NRM–TRM diagrams were analysed using the method of Kono & Tanaka (1984) and Williamson (1968).

Criteria for data analysis were based on the following four tests.

(1) Susceptibility test: room temperature initial magnetic susceptibility was measured with an MS-2 Bartington susceptibility meter after each step. The change in susceptibility after each step should be less than 10 per cent of the original susceptibility value.

(2) PTRM test: frequent checks were made on the reproducibility of partial thermoremanent magnetization (Coe, Gromme & Mankinen 1978). The PTRMs should be reproduced within errors contained in the TRM (Kono & Tanaka 1984).

(3) Linearity test: palaeointensity was determined on the NRM–TRM diagram from a linear segment consisting of more than four data points whose correlation coefficient exceeded 0.98. The linear segment was limited to the temperature range of the high-temperature component.

(4) Decay mode test: the decay mode of remanent magnetization intensity for a specimen in the palaeointensity experiment was compared with the decay mode of thermal demagnetization behaviour for another specimen for the same sample.

Typical successful examples are shown in Figs 4(a) to (d). The applied magnetic field was 30 μT for the specimen from GP010112 (Fig. 4a). In the temperature range 303 to 550 °C, the PTRM tests were successful and the magnetic susceptibility value after each step was similar to that before thermal demagnetization. Similar decay was observed in the remanent magnetization intensity for the palaeointensity experiment and

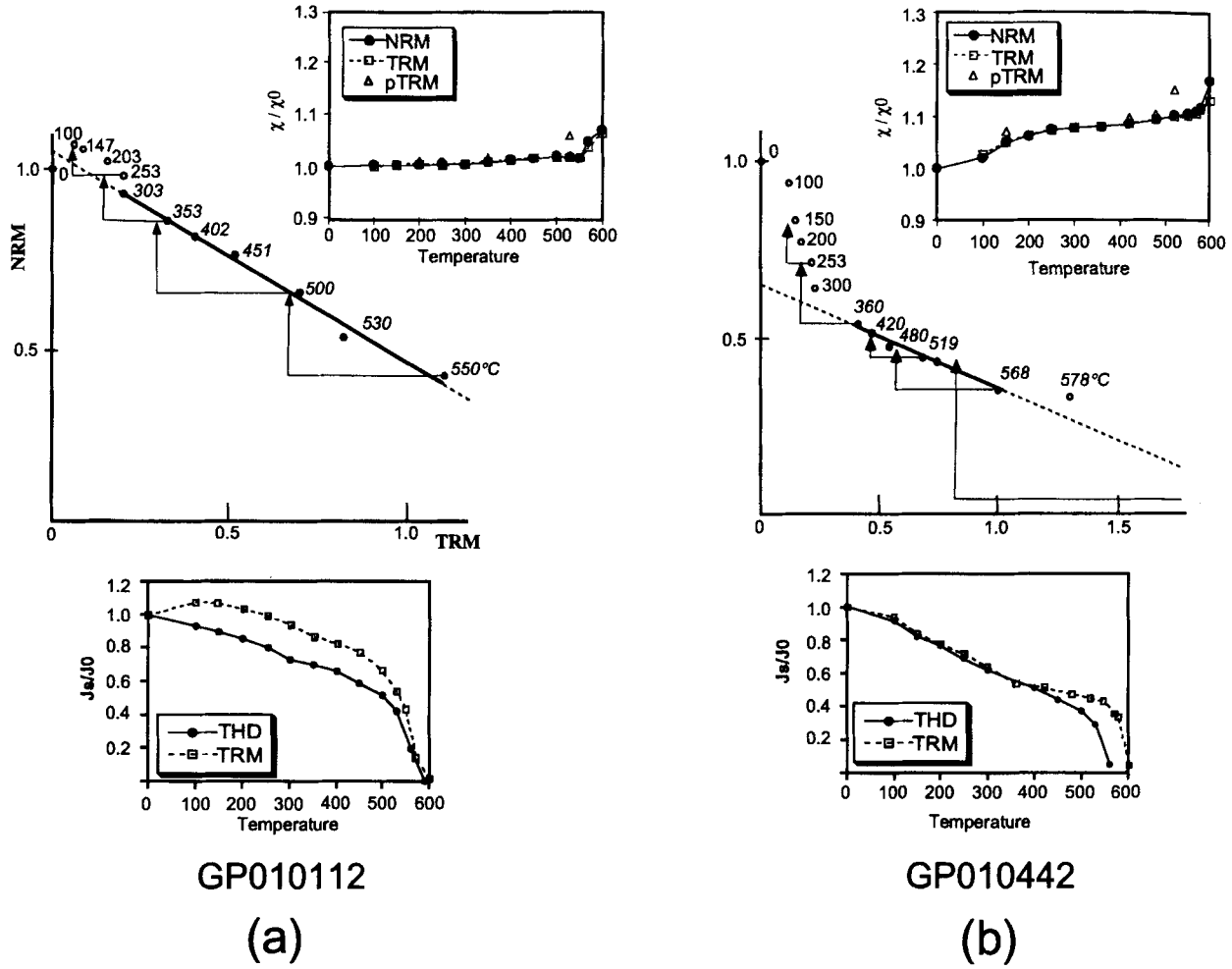


Figure 4. NRM–TRM diagrams for two typical successful examples [(a)–(d); GP010112, GP010442, GP010631 and GP010722] and one unsuccessful result [(e); GP011222]. Units of NRM and TRM axes are normalized by the NRM intensity. Closed (open) circles show the data used (rejected) for the linear fitting calculation. Numerals on the upper right are temperature steps in °C. The PTRM test is frequently applied: triangles show the points of the PTRM test with notation following that of Coe *et al.* (1978). Upper inset shows the change in magnetic susceptibility against the temperature step, and lower inset shows a comparison of the decay mode of the remanent magnetization intensity in palaeointensity experiments (solid line) and the thermal demagnetization behaviour (dashed line).

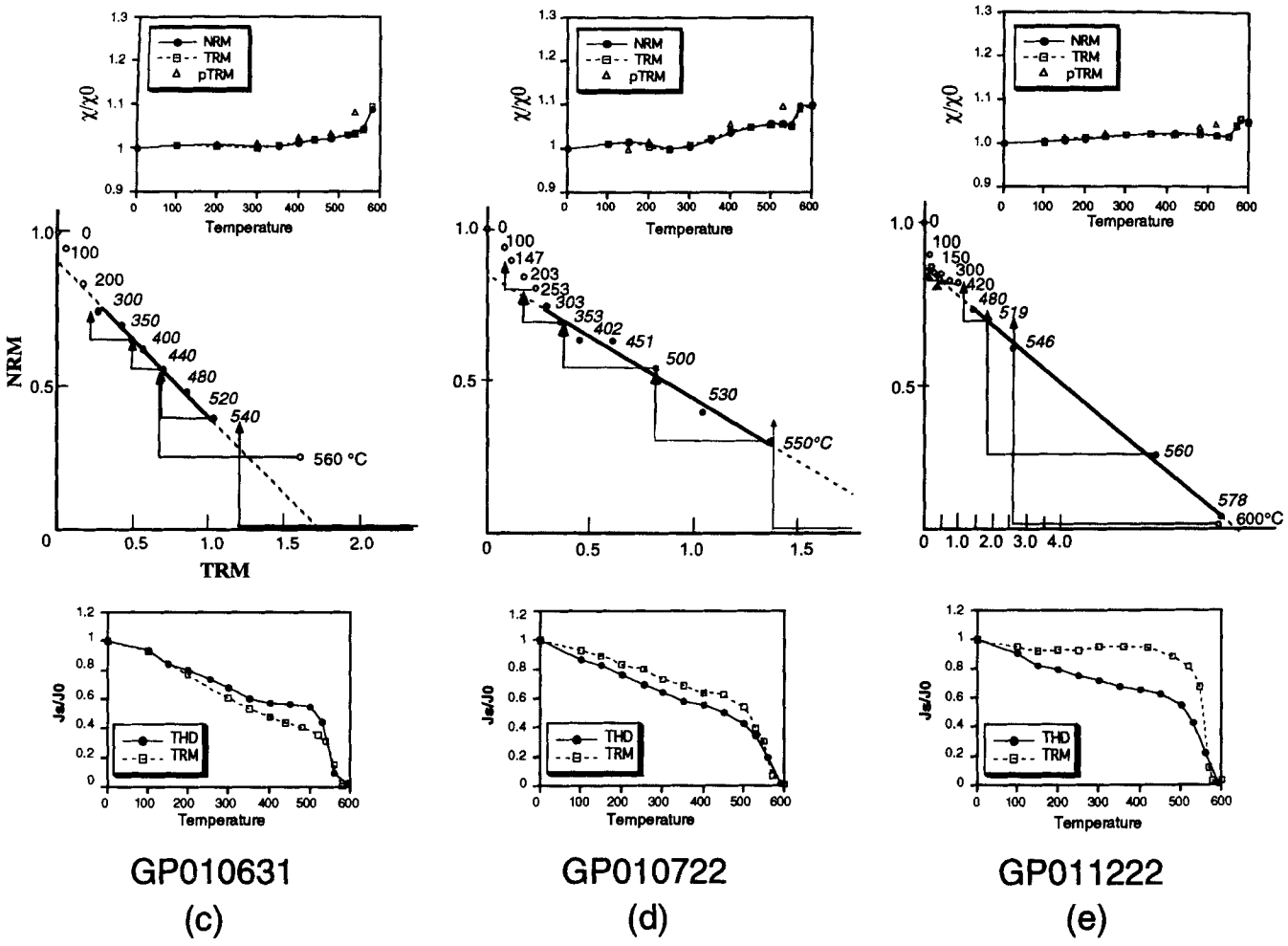


Figure 4. *(Continued.)*

Table 2. Palaeointensity data for successful results of the Nuuk dyke. Locality = distance of a sample from the contact; Jn = NRM intensity; T1–T2 = lowest and highest steps of the linear slope; n = number of points included in the linear regression; r = correlation coefficient of the linear NRM–TRM relationship; −b and sb = slope of the linear slope and its standard deviation; FL and Intensity = laboratory inducing field palaeointensity. The mean of all 13 data, and 12 data (sample GP010441 not used; denoted by an asterisk) are also listed.

Sample	Locality [m]	Specimen	Jn [mA/m]	T1-T2 [°C]	n	r	b	sb	FL [mT]	Intensity [mT]
GP0101	0.6	GP010112	10.7	303-550	7	0.9901	0.5904	0.0287	30	17.7
GP0102	0.2	GP010232	16.2	350-560	7	0.9913	0.3038	0.0194	25	7.6
GP0103	1.5	GP010322	15.3	350-555	7	0.9944	0.5948	0.0393	35	20.8
		GP010332	13.7	350-560	7	0.9901	0.4757	0.0224	25	11.9
GP0104	1.2	GP010441*	50.6	303-530	6	0.9954	1.4310	0.1254	30	42.9
		GP010442	24.4	360-568	6	0.9907	0.2851	0.0217	50	14.3
GP0106	1.3	GP010612	15.2	350-480	4	0.9954	0.6175	0.1109	35	21.6
		GP010631	13.5	300-540	7	0.9966	0.4927	0.0245	25	12.3
GP0107	1.1	GP010722	9.65	303-550	7	0.9944	0.4034	0.0161	30	12.1
GP0111	0.6	GP011121	24.9	440-540	4	0.9879	0.4689	0.0562	25	11.7
		GP011141	12.4	400-535	5	0.9853	0.3157	0.0274	35	11.1
GP0113	0.3	GP011322	17.9	250-519	7	0.9972	0.1668	0.0107	50	8.3
		GP011331	13.5	350-555	9	0.9986	0.5229	0.0180	25	13.1
mean					13					15.8 ± 9.2
mean*					12					13.5 ± 4.4

the thermal demagnetization. Seven data points between 303 and 550 °C define a straight line with a correlation coefficient of 0.9901. A palaeointensity estimate of 17.7 μT was obtained. The sample from GP010442 yields a palaeointensity of 14.3 μT under the condition of an applied magnetic field of 50 μT (Fig. 4b). The linear segment consisting of six data points above 360 °C has a correlation coefficient of 0.9907.

Unsuccessful results (sample GP011222) are illustrated in Fig. 4(e). A large difference is observed in the decay mode of remanent magnetization intensity, although the data points illustrate a linear relationship in the NRM–TRM diagram. Thermochemical remanent magnetization may have originated above 150 °C. The palaeointensity value for this specimen was rejected by the fourth test.

13 dyke samples pass the four tests (Table 2). Their palaeointensities range between 7.6 and 42.9 μT. These palaeointensities are not further corrected for the effects of cooling rate, because the dyke of width 4 m may have cooled over a short period of time (e.g. Turcotte & Schubert 1982).

The overall mean intensity and standard deviation for all data is 15.8 μT ± 9.2 μT. 12 intensity values in the 13 samples cluster in a narrow range between 7.6 and 21.6 μT, whereas one datum (42.9 μT) deviates from the mean by more than three times the standard deviation. The latter value was derived from a specimen whose NRM intensity is the strongest of all the specimens. The initial susceptibility value at 578 °C increases up to 120 per cent of the original value in this specimen, indicating that the anomalously strong TRM may be related to heating in the laboratory. No weight is, therefore, placed on this value, and the calculation based on the remaining 12 data determinations of 13.5 μT ± 4.4 μT is considered to give a more realistic estimate of the palaeointensity recorded by the dyke.

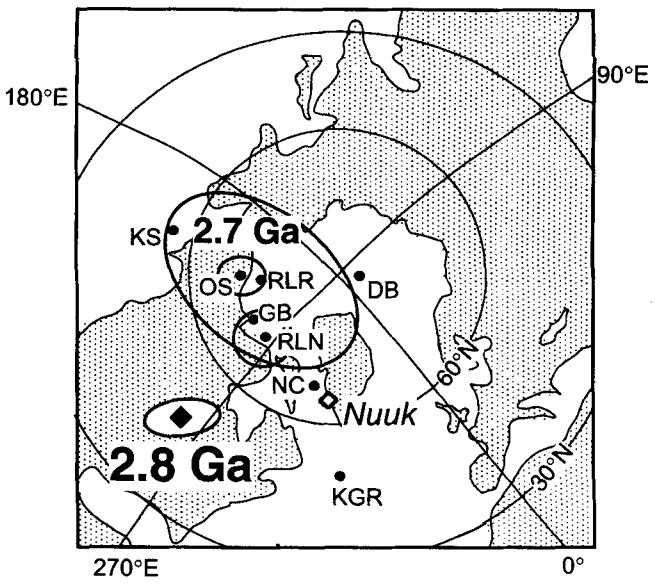


Figure 5. Palaeomagnetic pole for the 2.8 Ga dolerite dyke (shown by the closed diamond) and those of *ca.* 2.7 Ga rock units for North America, with their 95 per cent confidence limits. RLN and RNR: Red Lake greenstone with normal and reversed polarity, ~2700 Ma (Costanzo-Alvarez & Dunlop 1993); OS: Otto stock, 2680 Ma (Pullaiah & Irving 1975; Buchan, Neilson & Hale 1990; Corfu *et al.* 1989); DB, KGR, KS, GB and NC: Dobie Lake batholith, Kawashe Lake gabbro, Kwashe Lake Stock and North Caribu Lake batholith, ~2700 Ma (Hale & Lloyd 1989). Open diamond: present sampling locality (Nuuk). The 2.8 Ga palaeomagnetic pole position is shown after Greenland has been rotated through 18° clockwise back to North America using the rotation pole of Bullard *et al.* (1965).

DISCUSSION

The primary nature of the remanent magnetization of the dolerite dyke is suggested from a comparison of VGPs for the period 2.7 to 2.8 Ga as well as from the positive baked-contact test. The palaeomagnetic pole derived from the mean direction of the dyke is shown in Fig. 5. Since the dyke is intruded into the gneiss vertically, it is presumed that no significant tilting of the dyke has occurred. The palaeomagnetic pole position is shown after Greenland has been rotated through 18° clockwise back to North America about a Euler pole of 70.5°N, 94.4°W (Bullard, Everett & Smith 1965). The palaeomagnetic pole for the dyke is at 47.4°N, 258.7°E, and falls close to the Laurentian 2.7 Ga poles recently postulated by Costanzo-Alvarez & Dunlop (1993).

Low palaeointensity at 2.8 Ga has been documented by the present palaeomagnetic study. The mean palaeointensity (13.5 μT ± 4.4 μT) and the observed inclination value (74.6°) yield a virtual dipole moment (VDM) of $1.9 \pm 0.6 \times 10^{22}$ Am^2, assuming an unchanged Earth radius (McElhinny, Taylor & Stevenson 1978; Schmidt & Clark 1980). This VDM is about one-quarter of the present value.

The weak VDM at 2.8 Ga is presumably due to the long-term nature of the Earth's magnetism at about 2.7 ~ 2.8 Ga, although the palaeointensity of the 2.8 Ga dyke may represent only a spot-measure of the geomagnetic field because the dyke width is too thin to record geomagnetic secular change longer than about 10 yr (e.g. Turcotte & Schubert 1982). The VDM of the 2.8 Ga dolerite dyke is plotted on a graph of the variation of the VDM with time for the last 3.5 Ga constructed from the database of Tanaka & Kono (1994) in Fig. 6. Weak palaeointensity values have been reported for 2.7 Ga samples from the Stillwater Complex (Kobayashi 1968; Bergh 1970), the Pykes Hill komatiite, and Archaean basalt and gabbro (Hale 1987b). These correspond to a low magnitude of the VDM of between 2.6 and 5.2×10^{22} Am^2. The 2.8 Ga palaeointensity value from the present study is lower than the limits of dipole fluctuation observed for the past 2.0 Ga (Merrill & McElhinny 1983). This suggests that it is not simply ascribable to a temporary drop due to palaeosecular variation or geomagnetic dipole fluctuation.

The low-field period at 2.8 Ga appears to have been followed by a period of fairly strong geomagnetic field intensity, similar to the present value (Fig. 6): the Modipe gabbro (McElhinny & Evans 1968), Archaean diabase (Schwarz & Symons 1969) and the Matachewan diabase dyke (Kobayashi 1968; Hale 1987a) with an age of about 2.6 Ga revealed palaeointensities of 7.8 to 12.8×10^{22} Am^2. According to the compilation of mean VDMs with time of Merrill & McElhinny (1983), the Earth maintained a dipole moment close to the present-day value between 2.6 Ga and 0.7 Ga. Thus, a sudden increase in geomagnetic field intensity may have occurred in the period between 2.8 and 2.6 Ga.

The weak VDM value derived in the present study is supported by the value of $2.1 \pm 0.5 \times 10^{22}$ Am^2 reported for the 3.5 Ga Komati Formation in the Barberton Greenstone Belt (Hale 1987b) (Fig. 6). It should be noted that there is a

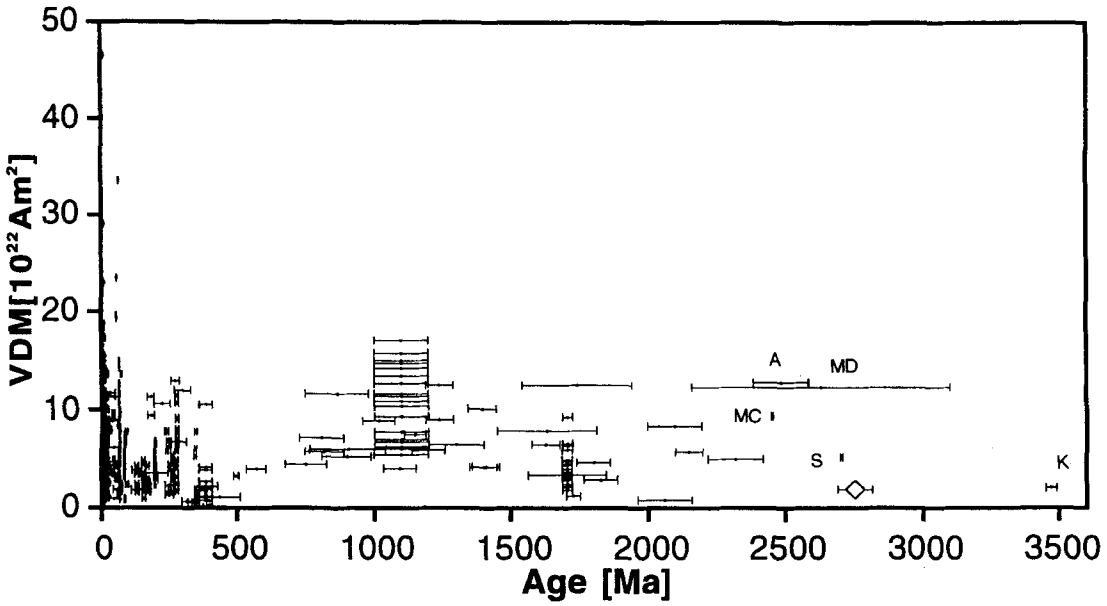


Figure 6. VDM for the 2.8 Ga dolerite dyke (open diamond), shown on a graph of the variation of VDM with time for the last 3.5 Ga constructed from the database of Tanaka & Kono (1994). K: Komati Formation in the Barberton Greenstone Belt (Hale 1987b); S: Stillwater Complex (Kobayashi 1968; Bergh 1970); MD: Modipe gabbro (McElhinny & Evans 1968); A: Archaean diabase (Schwarz & Symons 1969); MC: Matachewan diabase dyke (Kobayashi 1968; Hale 1987a).

long information gap between 3.5 and 2.8 Ga. More palaeointensity data from Archaean times are required, to establish whether low geomagnetic field intensities persisted between 3.5 and 2.8 Ga.

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